

SIMATS ENGINEERING ASSESSMENT

TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Dr.V.Parthipan

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Sivadharshini (192524778) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Sivadharshini

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than 1,200 employees use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company

Wireless Local Area Network (WLAN). The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use 4G/5G Wireless Wide Area Network (WWAN) services to access the company servers, customer databases, and business applications from remote locations. During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

Debugging & Optimization of WLAN and WWAN Infrastructure

1. Problem Analysis

The company is experiencing several wireless networking issues affecting productivity, communication, and security.

Problem	Possible Root Cause
Slow internet during peak hours	Network congestion, overloaded access points, insufficient bandwidth
Frequent Wi-Fi disconnections between floors	Poor roaming configuration, overlapping coverage, improper AP placement

Weak signals in conference rooms	Dead zones, building obstacles, inadequate AP coverage
Video conferencing freezes	High latency, packet loss, bandwidth saturation, interference
Uneven AP utilization	Poor load balancing and improper channel planning
WWAN connectivity loss in rural/highway areas	Weak cellular coverage, network handoff issues, limited carrier coverage
Outdated wireless security	Devices still using WEP/WPA instead of WPA3

2. Root Causes A. WLAN Issues

a) Network Congestion

- More than 1,200 users connect simultaneously.
- Cloud applications consume large bandwidth.
- Video conferencing generates heavy traffic.

Result

- Slow internet
- Increased latency
- Packet loss

b) Overloaded Access Points

Some APs serve many users while nearby APs remain lightly loaded.

Result

- Reduced throughput
- Higher response time
- Frequent disconnects

c) Poor Roaming

Employees move between six floors.

If roaming parameters are not configured properly,

- clients disconnect
- reconnect slowly
- meetings freeze

d) Poor Wireless Coverage

Conference rooms may contain

- concrete walls
- glass partitions
- electronic interference These weaken Wi-Fi signals.

e) Channel Interference

Nearby APs using the same Wi-Fi channel create

- co-channel interference
- adjacent channel interference This reduces throughput.

3. WWAN Issues

Weak Cellular Signal

Field engineers travel through

- rural areas
- highways
- remote locations

Some areas have weak 4G/5G coverage.

Result:

- upload failures
- delayed synchronization
- VPN disconnections

Poor Carrier Coverage

A single mobile operator may not provide good nationwide coverage.

Network Handover Problems

Moving vehicles frequently switch between

- cell towers
- LTE
- 5G

Improper handoff causes temporary loss of connectivity.

4. Security Problems

Audit found devices using

- WEP
- WPA

instead of

- WPA2
- WPA3

Risks include

- packet sniffing
- password cracking
- unauthorized access
- data theft

5. WLAN Optimization Techniques

A. Perform Wireless Site Survey

Use professional tools to identify

- dead zones
- interference
- signal strength
- AP placement

Outcome

- Better wireless coverage
- Improved signal quality

B. Install Additional Access Points

Deploy extra APs in

- conference rooms
- crowded workspaces
- cafeterias
- meeting halls
- Benefits
- Reduced congestion
- Better coverage
- Higher capacity

C. Load Balancing

Enable AP load balancing.

Instead of allowing one AP to serve 100 users, traffic is distributed across nearby APs.

Benefits:

- Better bandwidth utilization
- Faster connections
- Reduced overload

D. Band Steering

Move capable devices automatically to the

- 5 GHz
- 6 GHz (Wi-Fi 6E/7) band.

Leave only older devices on 2.4 GHz.

Benefits

- Less interference
- Higher speeds

E. Channel Optimization

Configure non-overlapping channels.

For example

2.4 GHz

- Channel 1
- Channel 6
- Channel 11

For 5 GHz

Use Dynamic Channel Assignment.

F. Quality of Service (QoS)

Prioritize

- Voice
- Video conferencing
- Business applications

over

- downloads
- software updates
- entertainment traffic

Benefits

- Smooth Zoom/Teams meetings
- Reduced latency

G. Fast Roaming

Enable

- IEEE 802.11r

- IEEE 802.11k
- IEEE 802.11v

Benefits

- Seamless roaming
- No call interruption
- Faster AP switching

6. WWAN Optimization

Multi-Carrier SIM

Use SIM cards supporting multiple mobile operators.

Benefits

- Automatic selection of strongest network
- Better rural connectivity

Signal Boosters

Install

- vehicle antennas
 - cellular boosters

for field engineers.

VPN Optimization

Use lightweight VPN protocols

- WireGuard ○ IKEv2

instead of slower legacy VPNs.

Offline Synchronization

Allow mobile applications to

- store data locally ○
synchronize
automatically
when the signal returns.

7. Security Improvements

Upgrade to WPA3

Replace all

- WEP ○ WPA

devices.

Benefits

- Strong encryption ○
Better authentication ○
Protection against
password attacks

Enable 802.1X Authentication

Use

- RADIUS Server ○
Enterprise Authentication

instead of shared passwords.

Network Segmentation

Separate

- Employee devices
- Guests
- IoT devices
- Printers using VLANs. Benefits
- Better security
- Reduced broadcast traffic

Regular Firmware Updates Update

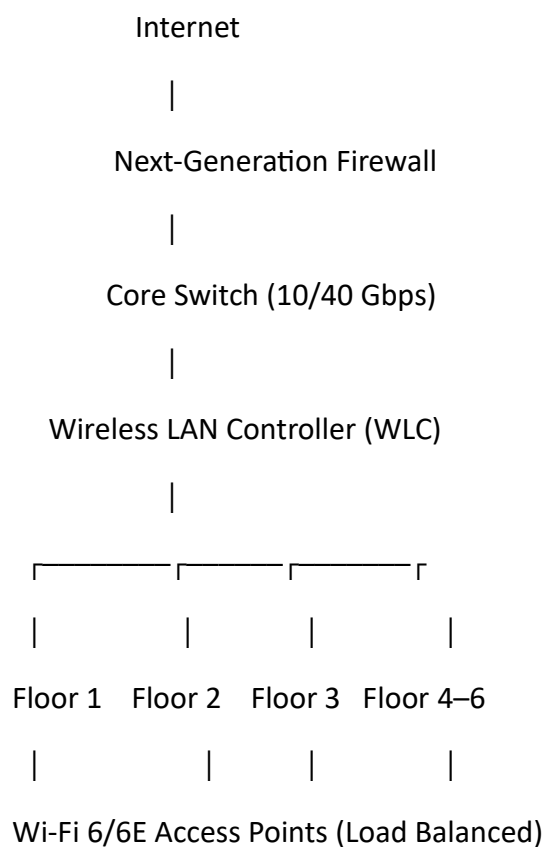
- Access Points
- Routers
- Firewalls
- Wireless Controllers to patch vulnerabilities.

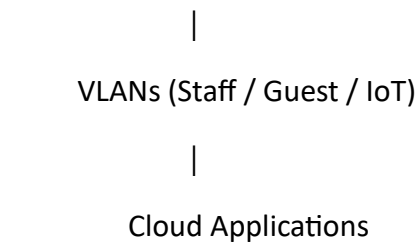
Continuous Monitoring

Deploy Wireless Intrusion Detection/Prevention Systems (WIDS/WIPS) to detect:

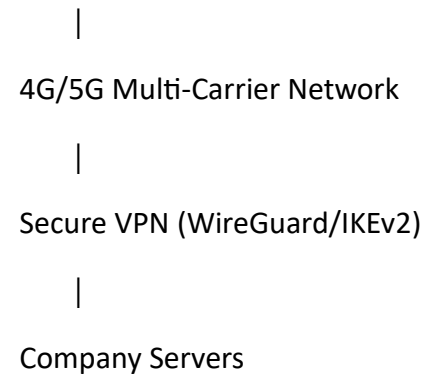
- Rogue access points
- Unauthorized devices
- Spoofing attacks
- Suspicious wireless activity

8. Recommended Network Design





Remote Employees



9. Expected Outcomes After Optimization

Issue	Optimization	Expected Result
Slow internet	Additional APs, QoS, load balancing	Faster network speeds
Wi-Fi disconnections	802.11r/k/v fast roaming	Seamless movement between floors
Weak conference room signals	Site survey and new AP placement	Strong, stable coverage
Video conferencing issues	QoS and channel optimization	Smooth HD video calls
Uneven AP usage	Load balancing	Balanced traffic distribution
WWAN signal loss	Multi-carrier SIMs, signal boosters, offline sync	More reliable remote connectivity
Security vulnerabilities	WPA3, 802.1X, VLANs, firmware updates	Stronger protection against cyber threats

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an Asynchronous Transfer Mode (ATM) network to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size 53-byte cells and establishes Virtual Circuits (VCs) to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

ATM Network Debugging & Optimization Report

1. Problem Restatement

Symptoms: transaction delays, VoIP jitter/loss, degraded video conferencing, high switch utilization, VC congestion, poor bandwidth allocation across traffic classes, and misuse of ATM service categories. This points to a **QoS/traffic-engineering fault**, not necessarily a raw capacity fault — the network has enough theoretical bandwidth, but it's being managed poorly.

2. Diagnostic Approach (Debugging Phase) Step 1 — Baseline the switches

- Check per-switch and per-port utilization during peak hours using SNMP/ATM MIB counters (cell discard rate, CLP-tagged cells, buffer occupancy).
- Identify which switches are consistently >70–80% utilized — these are congestion hotspots.

Step 2 — Audit VC/VP configuration

- List all PVCs/SVCs and their assigned **ATM service category** (CBR, rt-VBR, nrtVBR, ABR, UBR).
- Cross-check against the *actual* application running on each VC. This is almost certainly where the core fault lies.

Step 3 — Identify service-category mismatches

Common real-world misconfigurations that produce exactly these symptoms:

Application	Should Use	Likely Misconfigured As	Effect
VoIP	CBR or rt-VBR	UBR or nrt-VBR	Jitter, delay, dropped cells
Video conferencing	rt-VBR	UBR / ABR	Variable quality, frame loss
Online banking / transactions	nrt-VBR or ABR	UBR (best-effort)	Delay under congestion, transaction timeouts
Bulk data/backup	UBR or ABR	(fine as-is)	—

If real-time voice/video traffic is riding on UBR (no guarantees, no policing priority), it will always suffer first when the network is busy — this alone explains the jitter, loss, and video quality complaints.

Step 4 — Check traffic contracts and policing (UPC)

- Verify PCR (Peak Cell Rate), SCR (Sustainable Cell Rate), MBS (Maximum Burst Size), and CDVT (Cell Delay Variation Tolerance) parameters for each VC.
- Look for traffic contract violations — cells tagged CLP=1 or discarded due to non-conforming traffic (bursty video/data exceeding SCR).
- Under-provisioned SCR/PCR values force legitimate traffic to be marked nonconformant, causing unnecessary drops. **Step 5 — Examine buffer/queue management**
- Check if switches use simple FIFO queuing across all classes instead of perclass queues with weighted scheduling — this causes real-time traffic to sit behind bulk/data traffic.
- Check for buffer overflow patterns correlating with peak hours (indicates insufficient CAC — Connection Admission Control — enforcement). **Step 6 — Verify Connection Admission Control (CAC)**
- If CAC is too permissive, more VCs are admitted onto a link than its committed bandwidth can support at guaranteed rates → oversubscription → congestion during peak load.

3. Root Causes Identified

1. **Service category mismatch** — real-time traffic (VoIP, video) placed on UBR/nrtVBR instead of CBR/rt-VBR.

2. **Poor CAC enforcement** — too many VCs admitted without checking aggregate bandwidth/QoS guarantees, causing oversubscription on hot links.
3. **Inadequate traffic shaping/policing parameters** — PCR/SCR set too low for actual traffic bursts, causing cell tagging/discard.
4. **FIFO-only queuing** at congested switches instead of priority-based scheduling per VC class.
5. **Static, non-adaptive bandwidth allocation** — no dynamic reallocation between under- and over-utilized VCs.

4. Optimization & Fix Recommendations

A. Reassign correct ATM Service Categories

- **VoIP** → CBR (fixed low-bandwidth, low-jitter guarantee) or rt-VBR if some burst tolerance is acceptable.
- **Video conferencing** → rt-VBR (handles bursty but delay-sensitive traffic with bounded jitter).
- **Online banking/transaction processing** → nrt-VBR (guarantees average throughput without strict jitter needs) or ABR with a reasonably high MCR (Minimum Cell Rate).
- **Bulk/batch data, backups, non-critical traffic** → UBR (best-effort, lowest priority).

B. Tune traffic contracts

- Recompute PCR/SCR/MBS based on actual measured traffic profiles (use monitoring data from Step 1–2) rather than default/legacy values.
- Add adequate CDVT to avoid unnecessary cell tagging on jitter-sensitive but slightly bursty flows (video).

C. Strengthen CAC policy

- Enforce strict admission control: reject/renege new VCs if aggregate CBR+rtVBR guaranteed bandwidth would exceed a safe threshold (e.g., 80–85%) of link capacity.
- Reserve dedicated bandwidth pools per traffic class on trunk links between HQ, regional offices, and data centers.

D. Improve queuing/scheduling

- Implement per-class priority queuing at each switch: CBR served first (low latency), rt-VBR next, then nrt-VBR/ABR, with UBR last.
- Use Weighted Fair Queuing (WFQ) or similar weighted scheduling between nrtVBR/ABR/UBR to prevent starvation while still prioritizing real-time classes.

E. Load balancing and capacity planning

- Redistribute VCs across underutilized switches/links (traffic engineering) rather than concentrating financial-transaction and voice/video VCs on the same congested trunks.
- Add capacity (upgrade trunk speed or add parallel VPs) on links consistently running >80% utilization at peak.

F. Monitoring and feedback loop

- Deploy continuous ATM MIB/SNMP monitoring with alerting on cell loss ratio (CLR), cell delay variation (CDV), and cell transfer delay (CTD) per VC.
- Set up periodic (e.g., weekly) review of traffic contracts vs. actual usage to catch drift before it causes visible service degradation.

5. Expected Outcome

Metric	Before	After Optimization
VoIP jitter/loss	High	Minimal (CBR/rt-VBR guarantees)
Video quality	Degraded	Stable (rt-VBR with correct SCR/MBS)
Transaction delay	Noticeable	Reduced (nrt-VBR/ABR with proper CAC)
Switch utilization	Uneven, hotspots	Balanced via traffic engineering
Bandwidth efficiency	Poor (misallocated)	Improved (class-based allocation matches actual demand)

Summary: The core fault isn't lack of bandwidth — it's that real-time and transaction traffic isn't classified or scheduled correctly. Correcting service-category assignment, tightening CAC, tuning traffic contracts, and adding priority-based queuing should resolve the reported jitter, delay, and quality issues without necessarily requiring new hardware.

3.A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a Multiprotocol Label Switching (MPLS) network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also

discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

MPLS Network Debugging & Optimization Report

1. Problem Restatement

Symptoms: slow cloud app access, delayed order processing, poor VoIP quality, video conferencing interruptions, congested LSPs alongside underutilized paths, incorrect label mappings, and absent/misconfigured traffic engineering. This is primarily a **control-plane and traffic-engineering fault**, not a raw bandwidth shortage — traffic is being funneled onto a few LSPs while other paths sit idle, and some routers are mislabeling/misrouting packets.

2. Diagnostic Approach (Debugging Phase) Step 1 — Verify LSP status and utilization

- Check LSP state on all LERs (Label Edge Routers) and LSRs (Label Switch Routers):
show mpls lsp, show mpls forwarding-table (or vendor equivalent).
- Correlate per-LSP utilization with peak-hour complaints — identify which LSPs are >80% utilized versus near-idle ones.

Step 2 — Audit the Label Information Base (LIB) / Label Forwarding Information Base (LFIB)

- Compare label bindings across adjacent routers (show mpls forwarding-table, LDP/RSVP-TE bindings) to catch inconsistent or stale label mappings.
- Look for label mismatches caused by:
 - LDP session flaps/resyncs leaving stale entries
 - Manual/static label misconfiguration
 - MTU mismatches causing fragmented label-switched packets
- Trace a few affected flows end-to-end (label imposition at ingress → swap at each hop → pop at egress) to confirm where mismapping occurs. **Step 3 — Check signaling protocol health**
- If using LDP: verify LDP neighbor sessions are stable (no repeated flapping causing label withdrawal/re-advertisement).

- If using RSVP-TE: check for path re-signaling failures, PATH/RESV message errors, and whether explicit routes (ERO) are being honored. **Step 4 — Examine Traffic Engineering (TE) configuration**
- Check whether **MPLS-TE / RSVP-TE tunnels** exist at all, or if traffic is following default IGP shortest-path (which causes exactly this pattern: some links overloaded, others idle).
- Verify whether **CSPF (Constrained Shortest Path First)** is enabled to route LSPs based on available bandwidth rather than pure hop count.
- Check if bandwidth reservations per LSP were ever configured, and if link-state (IGP-TE extensions: OSPF-TE or ISIS-TE) is properly advertising available bandwidth.

Step 5 — Review QoS / traffic classification

- Confirm each application is mapped to the correct **Traffic Class (EXP bits / TC field)**:
 - VoIP/video → Expedited Forwarding (EF)
 - ERP/order processing/transactions → Assured Forwarding (AF) with appropriate drop precedence
 - Bulk/cloud sync/non-critical → Best Effort
- Look for misclassification causing real-time traffic to compete with bulk traffic in the same queue.

Step 6 — Check load balancing / ECMP behavior

- Determine if multiple equal-cost LSPs exist but traffic isn't being distributed across them (common when hashing is based on limited fields, or ECMP isn't enabled for MPLS).

3. Root Causes Identified

1. **No/ineffective traffic engineering** — LSPs following IGP shortest path rather than constraint-based (CSPF) routing, causing some links to be overloaded while parallel paths sit idle.
2. **Incorrect/stale label mappings** — due to LDP session instability, manual misconfiguration, or unsynchronized LFIBs across routers, causing misrouted or looped traffic and added latency.
3. **Missing bandwidth reservation** — LSPs not signaled with proper bandwidth constraints, so CAC-like protection doesn't exist; congestion-prone links keep getting more LSPs mapped onto them.

4. **No/weak QoS classification** — real-time voice/video traffic not marked EF, competing with bulk data in the same queue during congestion.
5. **Static routing without failover/load balancing** — no use of ECMP or autobandwidth adjustment to redistribute traffic from congested to underutilized LSPs.

4. Optimization & Fix Recommendations

A. Fix label mapping errors

- Resynchronize LDP sessions across all routers; clear stale label bindings (clear mpls ldp neighbor / equivalent).
- Audit static label configurations for typos/mismatches; migrate to dynamic LDP or RSVP-TE where manual config is error-prone.
- Verify consistent MTU across the LSP path to prevent fragmentation-related issues.

B. Implement proper MPLS Traffic Engineering

- Enable **RSVP-TE with CSPF** so LSPs are placed based on actual available bandwidth, not just shortest IGP path.
- Advertise link bandwidth via **OSPF-TE / ISIS-TE** extensions so CSPF has accurate topology + bandwidth data.
- Configure explicit bandwidth reservations per LSP matching real application requirements (VoIP, video, ERP, bulk).

C. Enable load balancing across paths

- Configure **multiple parallel LSPs** between key site pairs (HQ–cloud DC, warehouse–regional office) with **load balancing / traffic splitting** across them.
- Use **auto-bandwidth** (RSVP-TE auto-bw) so LSP reserved bandwidth adjusts dynamically to actual traffic demand, preventing over/under-provisioning.

D. Apply LSP prioritization and preemption

- Assign **setup/holding priorities** to LSPs so critical traffic (VoIP, order processing) can preempt lower-priority LSPs (bulk data) during congestion.
- Configure **Fast Reroute (FRR)** for critical LSPs to provide sub-50ms failover on link/node failure, avoiding voice/video drops during path changes.

E. Correct QoS/traffic classification

- Map applications to correct MPLS EXP/TC values:
 - VoIP/video conferencing → EF (highest priority, low latency queue)

- Order processing, ERP, customer support → AF with appropriate drop precedence
- Bulk cloud sync, backups → Best Effort
- Implement per-hop scheduling (Priority Queuing + WFQ) consistent with these classes at every LSR.

F. Continuous monitoring and verification

- Deploy MPLS OAM tools: **LSP ping** and **LSP traceroute** to regularly verify label path integrity and detect misrouting/loops proactively.
- Monitor per-LSP bandwidth utilization, packet loss, and delay continuously; set alerts for LSPs exceeding 75–80% utilization.
- Periodically audit label bindings across the network to catch drift before it causes visible faults.

5. Expected Outcome

Metric	Before	After Optimization
LSP utilization	Uneven (some congested, some idle)	Balanced via CSPF + load balancing
Label mapping errors	Present, causing misroutes	Resolved via LDP resync + audits
VoIP/video quality	Poor (jitter, drops)	Improved (EF class + FRR + priority)
Order processing delay	High during peak	Reduced (AF class + bandwidth-guaranteed LSPs)
Path failure recovery	Slow/manual	Sub-50ms via Fast Reroute
Bandwidth efficiency	Poor allocation	Optimized via TE + auto-bandwidth

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical

interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

Warehouse Network — Transmission Error Debugging & Optimization Report

1. Problem Restatement

Symptoms: corrupted shipment records, failed barcode scans, inventory mismatches, rising transmission errors during peak hours (electrical interference, attenuation, congestion), and — critically — **some corrupted frames are being retransmitted correctly while others go undetected and silently corrupt the database**. This last point is the key clue: the fault lies in the **error detection/correction mechanism itself being inadequate or misconfigured**, not just noisy physical links.

2. Diagnostic Approach (Debugging Phase)

Step 1 — Characterize the physical layer noise

- Identify sources of electrical interference: motors on autonomous robots, conveyor systems, forklift chargers, fluorescent/LED ballasts, and RF interference from many RFID readers/Wi-Fi APs operating in the same band.
- Measure signal attenuation on cabling runs (especially long warehouse-floor Ethernet/coax runs) using a cable tester/TDR — check against category rating limits (e.g., Cat5e/6 100m limit).
- Check Wi-Fi channel overlap/co-channel interference if scanners/robots use wireless (2.4GHz especially, since it's crowded and interference-prone).

Step 2 — Identify which error-detection scheme is in use

- Check current mechanism at data-link layer: is it a **simple parity check, checksum, or CRC (Cyclic Redundancy Check)**?
- Parity checks only detect odd numbers of bit errors and are **weak against burst errors** (very likely given electrical interference causes burst noise, not isolated single-bit flips).
- If parity or a weak checksum is being used, this directly explains why **some corrupted frames go undetected** — burst errors easily fool single-bit parity.

Step 3 — Examine retransmission (ARQ) behavior

- Check the ARQ scheme (Stop-and-Wait, Go-Back-N, Selective Repeat) — is retransmission timeout appropriately tuned for peak-hour congestion delay?
- If timeouts are too short, unnecessary retransmissions add to congestion; if too long, real errors sit undetected/unresolved longer.
- Verify whether error detection is even being applied consistently across all device types (barcode scanners, RFID readers, robots) — often the weakest-link device (e.g., legacy scanners) uses outdated or no CRC. **Step 4 — Correlate errors with peak-hour congestion**
- Check if buffer overflows/collisions during peak hours are causing frame drops that get misinterpreted as "successful" transmissions at the application layer (i.e., error detection is skipped or ignored due to protocol misconfiguration, not just insufficient).
- Check whether corrupted frames are being detected at the link layer but the application layer doesn't reconcile/re-request data, allowing corrupted data through to the database anyway (a protocol-layering gap, not just a physical error rate problem).

Step 5 — Audit database write logic

- Determine if inventory/shipment updates are applied even when a frame fails checksum (e.g., partial writes, no application-level validation) — this indicates the fault isn't purely physical/data-link, but also a lack of end-to-end validation before committing to the database.

3. Root Causes Identified

1. **Weak error-detection code** — likely simple parity or a weak checksum, which cannot reliably catch burst errors typical of electrical interference (parity fails on even-numbered bit flips within a byte; burst noise commonly flips multiple bits together).
2. **No/inadequate CRC implementation** — CRC (e.g., CRC-16/CRC-32) is far more robust against burst errors and is the industry standard for this exact scenario, but may not be properly enabled on all device types.
3. **Inconsistent error control across device types** — legacy barcode scanners or RFID readers may not implement the same detection strength as newer devices, creating gaps.
4. **Physical layer degradation** — attenuation from long cable runs and EMI from warehouse machinery are increasing raw bit-error rate beyond what current detection can reliably manage.

5. **Congestion-induced frame loss** — peak-hour congestion causing buffer overflows/collisions, compounding the error rate on top of physical noise.
6. **Missing end-to-end/application-layer validation** — corrupted-but-undetected frames are being committed straight to the database with no reconciliation check.

4. Optimization & Fix Recommendations

A. Upgrade error-detection mechanism

- Replace parity/weak checksum with **CRC-32** (or at minimum CRC-16) at the data-link layer — CRC is specifically designed to catch burst errors, which matches the EMI-driven error pattern described.
- Ensure CRC is applied **uniformly** across all device classes (scanners, RFID readers, robot controllers) — standardize firmware/drivers if legacy devices lack it.

B. Strengthen error correction where retransmission isn't ideal

- For latency-sensitive real-time data (robot control commands), consider **Forward Error Correction (FEC)** (e.g., Hamming code or Reed-Solomon) so minor errors can be corrected without waiting for retransmission — reduces delay-sensitive failures.
- For non-real-time data (shipment/inventory records), continue with **ARQ (Automatic Repeat reQuest)** but tune retransmission timers based on measured peak-hour round-trip delay.

C. Improve ARQ efficiency

- Move from Stop-and-Wait (if in use) to **Selective Repeat ARQ** to avoid retransmitting entire blocks of frames unnecessarily, improving throughput during congestion.
- Set adaptive timeout values that scale with current network congestion rather than a fixed timer.

D. Reduce physical layer error sources

- Replace/shield cabling in high-EMI zones (near motors, conveyors) with shielded twisted pair (STP) or migrate to fiber where feasible.
- Verify cable run lengths are within specification; add repeaters/switches to segment long runs.
- For wireless scanners/RFID, perform a site survey and reassign non-overlapping channels (or migrate to 5GHz/Wi-Fi 6 where interference is lower) and increase RFID reader spacing/power tuning to reduce mutual interference.

E. Reduce congestion contribution to errors

- Segment the warehouse network (VLANs) by function — separate scanner/RFID traffic, robot control traffic, and inventory server traffic — to reduce collision domains and contention.
- Apply QoS prioritization for time-critical robot control and scan data over bulk inventory sync traffic.
- Consider upgrading peak-hour bottleneck links (switch uplinks, AP backhaul) if utilization consistently exceeds safe thresholds.

F. Add end-to-end validation before database commit

- Implement **application-layer checksums/hashes** on shipment/inventory records so even if a data-link error slips through, the application can detect and reject the corrupted record before it's written to the database.
- Add automated reconciliation jobs that periodically cross-check database inventory counts against physical stock counts (cycle counting) to catch and correct residual undetected errors quickly.

G. Monitoring and continuous verification

- Deploy frame-error-rate monitoring per segment/device to catch degrading cable runs or interference sources before they cause visible data corruption.
- Track CRC failure rate, retransmission rate, and database reconciliation mismatch rate as ongoing KPIs, with alerting thresholds for peak-hour degradation.

5. Expected Outcome

Metric	Before	After Optimization
Undetected corrupted frames	Frequent (weak parity/checksum)	Rare (CRC-32 catches burst errors)
Inventory/database mismatches	Frequent	Significantly reduced (CRC + app-layer validation)
Barcode scan failures	Elevated during peak hours	Reduced (EMI mitigation + CRC)
Retransmission efficiency	Poor (possible Stopand-Wait)	Improved (Selective Repeat + adaptive timers)
Real-time robot control errors	Delay-prone (pure ARQ)	Improved (FEC reduces retransmission need)
Network congestion impact on errors	High	Reduced via VLAN segmentation + QoS

5. A leading online education platform conducts live classes and online examinations for more than 15,000 students across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed 10 Gbps connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

Online Education Platform — Flow Control Debugging & Optimization Report

1. Problem Restatement

Symptoms: incomplete downloads, delayed page loads, duplicate packets, excessive retransmissions, receiver buffer overflow, application crashes, increased latency, reduced throughput, poor bandwidth utilization. The core mismatch: a **10 Gbps server** is sending data to thousands of clients on much slower, heterogeneous links (broadband/mobile). This is a textbook **sender-receiver speed mismatch / flow control failure** — the server is transmitting faster than individual receivers (and the network path to them) can consume, and the current flow control mechanism isn't throttling the sender appropriately per-receiver.

2. Diagnostic Approach (Debugging Phase)

Step 1 — Identify the flow control scheme in use

- Check whether the transport layer is using **fixed-size window flow control** or a properly adaptive one (TCP sliding window with dynamic receive window advertisement).
- A fixed/static window (or a window that doesn't shrink to match slow client throughput) explains why fast broadband/mobile mismatches cause overflow — the sender keeps pushing data sized for its own capacity, not the receiver's.

Step 2 — Confirm the buffer overflow symptom

- "Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow" is a **direct flowcontrol failure signature**: the receiver's advertised window (rwnd) either isn't being honored, isn't being updated fast enough, or the application isn't reading from the socket buffer in time (slow consumer).
- Check if receiver window updates (ACKs carrying updated rwnd) are being processed by the sender, or if the sender is ignoring/overriding them (e.g., misconfigured send buffer that ships data regardless of rwnd = 0 signals).

Step 3 — Check for Silly Window Syndrome and related inefficiencies

- Look for evidence of **Silly Window Syndrome (SWS)**: receiver repeatedly advertising small windows as it slowly drains a full buffer, sender sending small segments in response — this increases overhead and worsens throughput on slow clients.
- Verify **Nagle's algorithm** and **delayed ACK** settings — a bad interaction between the two can cause stalls (each waiting on the other), especially over higherlatency mobile links.

Step 4 — Examine congestion control interaction

- Even though this is fundamentally a flow-control issue, check congestion control behavior too: is the server using a congestion window (cwnd) that's too aggressive for mobile/broadband RTTs, causing packet loss → retransmissions → duplicate packets at the receiver (which matches the reported duplicate-packet symptom)?
- Duplicate packets strongly suggest retransmission timers firing before ACKs are received (RTT misestimation), not just genuine loss — check if RTO (Retransmission Timeout) values are properly adaptive (Karn's algorithm, RTT

variance) or statically set too short for high-latency mobile paths. **Step 5 — Check for one-size-fits-all transmission**

- Confirm whether the server treats all 15,000+ students identically (same window size, same segment size, same pacing) regardless of their access link type (10 Gbps campus vs. 3G mobile vs. home broadband) — this uniform treatment is very likely a major root cause given the described heterogeneity. **Step 6 — Review multicast/broadcast delivery method**
- Since the server "simultaneously sends question papers, images, videos... to all students," check if this is unicast-per-student (scales poorly, 15,000 individual flows competing for the 10 Gbps uplink and amplifying congestion) versus a more scalable delivery method.

3. Root Causes Identified

1. **Speed/capacity mismatch between sender and receivers** — 10 Gbps server transmitting without adequately adapting to much slower, heterogeneous client links.
2. **Ineffective/static flow control window** — sender not properly honoring or adapting to each receiver's advertised window, leading to overflow when slow clients can't drain data fast enough.
3. **Silly Window Syndrome** — small, inefficient window updates/segments during buffer-constrained conditions, worsening throughput.
4. **Poorly tuned retransmission timers (RTO)** — causing premature retransmissions and duplicate packets, especially over higher-latency mobile connections.
5. **Uniform transmission strategy** — same treatment for all 15,000 students regardless of actual link capacity/RTT, ignoring client diversity.
6. **Congestion control not properly cooperating with flow control** — aggressive sending overwhelming both the network path and the receiver simultaneously during peak exam hours.
7. **Inefficient content delivery model** — likely large-scale unicast delivery of identical content (question papers/videos) to thousands of clients simultaneously, unnecessarily multiplying load.

4. Optimization & Fix Recommendations

A. Fix flow control mechanism

- Ensure proper **dynamic sliding window flow control** is enabled and strictly honored — sender must throttle to the receiver's currently advertised window (rwnd), not just its own send capacity.
- Enable/verify **window scaling (RFC 1323)** so large windows can be advertised for high-bandwidth paths, while still shrinking appropriately for constrained mobile clients.
- Implement or verify protection against **Silly Window Syndrome**: use Clark's solution (receiver delays window update until meaningful space is free) and Nagle's algorithm together correctly (not both delaying simultaneously and causing stalls).

B. Tune retransmission and timeout behavior

- Implement **adaptive RTO calculation** (based on measured RTT and variance, per-connection) rather than a fixed timeout, so mobile (high-RTT) clients don't trigger premature retransmissions/duplicates.
- Apply **Karn's algorithm** to avoid corrupting RTT estimates with retransmitted segment timings.

C. Adapt transmission per client profile

- Implement **per-connection congestion/flow adaptation**: detect each client's effective throughput/RTT early (e.g., via TCP slow start behavior or explicit bandwidth probing) and adjust segment size (MSS) and pacing accordingly.
- Consider **adaptive bitrate / adaptive content delivery** for video content specifically — serve lower-resolution video/images to detected low-bandwidth clients instead of pushing full-quality content uniformly.

D. Reduce redundant/simultaneous load

- Move from per-student unicast delivery of identical static content (question papers, images, exam videos) to a **CDN (Content Delivery Network) or caching/edge server** model — static exam materials should be cached close to students, not re-sent individually from the central server each time.
- For live/simultaneous delivery components, evaluate **multicast delivery** where network support allows, to avoid 15,000 duplicate unicast streams competing for the same uplink.

E. Apply server-side rate control and traffic shaping

- Implement **rate limiting/traffic shaping per connection** at the server or load balancer so no single fast burst overwhelms the aggregate network path or individual slow clients.
- Use a **load balancer / reverse proxy tier** (rather than one central 10 Gbps server handling 15,000 direct connections) to distribute connection handling and apply consistent, controlled pacing to each client based on its measured capacity.

F. Application-layer buffering safeguards

- Ensure client applications acknowledge processed data and request next content chunk explicitly (**pull-based delivery**, e.g., adaptive chunked download) rather than the server pushing subsequent data blindly — this directly prevents "next data arriving before device finishes processing previous."
- Add server-side pacing that waits for acknowledgment/buffer-availability signals before sending the next batch, rather than relying purely on transport-layer flow control to catch overflow after the fact.

G. Monitoring and load testing

- Conduct **peak-load simulation testing** (15,000+ concurrent simulated clients with varied bandwidth/RTT profiles) before exam days to validate flow control tuning holds up under real conditions.

- Monitor real-time per-connection metrics during exams: retransmission rate, duplicate ACK rate, receive window size trends, and buffer occupancy — with alerting thresholds to catch degradation early.

5. Expected Outcome

Metric	Before	After Optimization
Receiver buffer overflow / crashes	Frequent	Rare (proper window honoring + pull-based delivery)
Duplicate packets / retransmissions	Excessive	Significantly reduced (adaptive RTO + Karn's algorithm)
Throughput for slow clients	Poor	Improved (adaptive segment size/content quality)
Server load / bandwidth utilization	Inefficient (uniform push to all)	Optimized (CDN/caching + perclient pacing)
Page load / download completion	Delayed, incomplete	Consistent, complete
Scalability to 15,000+ concurrent students	Strained	Improved (load balancing + edge distribution)